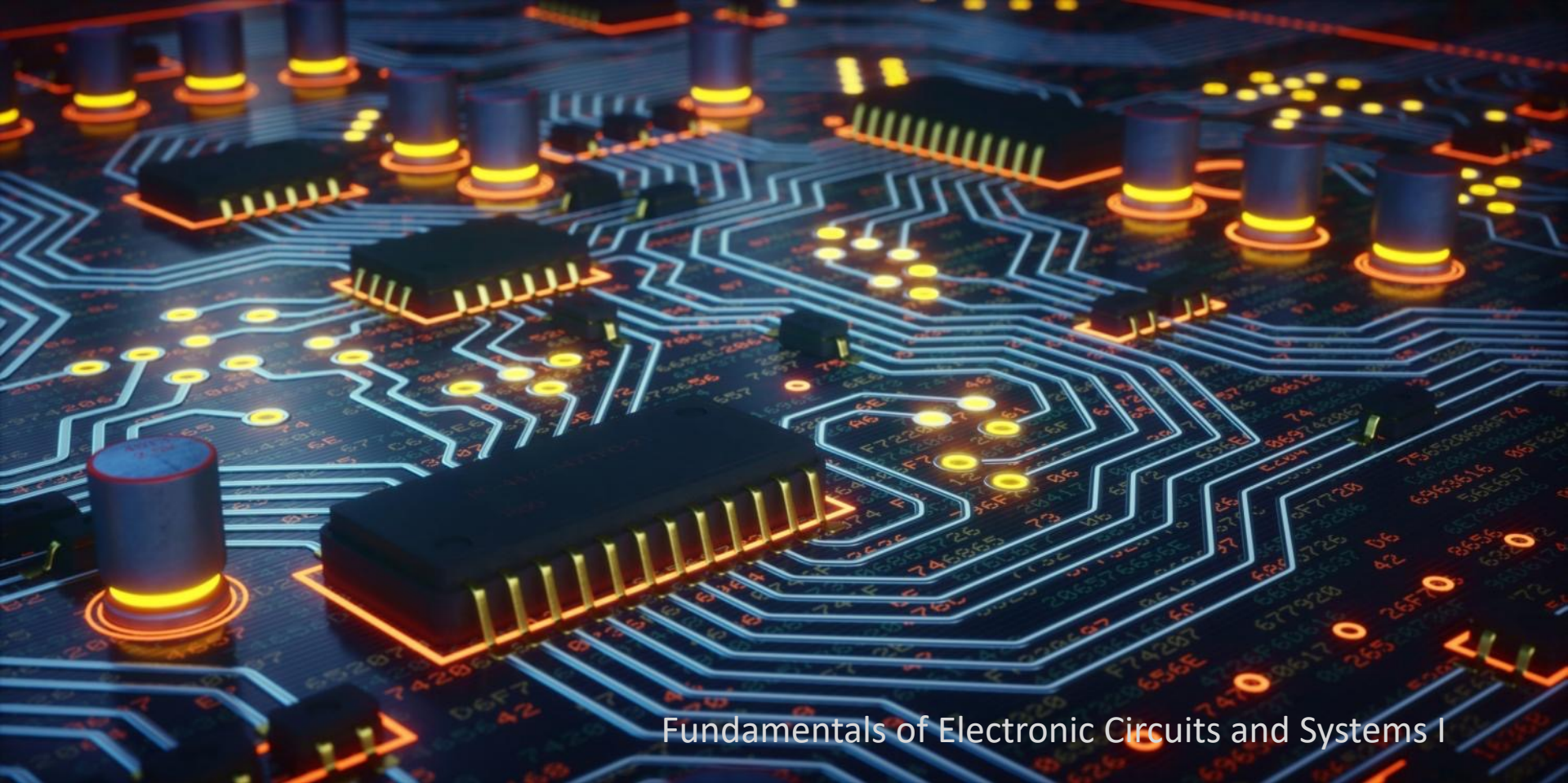




Fundamentals of Electronic Circuits and Systems I

Introduction & Basic Concept

Milin Zhang

Dept of EE, Tsinghua University



Welcome

- Instructor: [Milin Zhang](#)
 - Email: zhangmilin [at] tsinghua [dot] edu [dot] cn
 - Office location: Rm713, Rohm BLDG
 - Office hour: by email appointment
- TAs
 - [Tianyi Liu](#) (liuty21[at] mails [dot] tsinghua [dot] edu [dot] cn)
 - [Jiaxin Lei](#) (leijx19[at] mails [dot] tsinghua [dot] edu [dot] cn)
 - [Yahao Song](#) (songyh21[at] mails [dot] tsinghua [dot] edu [dot] cn)
 - [Yusong Wu](#) (wys22 [at] mails [dot] tsinghua [dot] edu [dot] cn)
 - Office hour: Sat 7-10pm, Rm 715, Rohm BLDG
- Textbook
 - “[Basic Engineering Circuit Analysis](#)” by J. D. Irwin & R. M. Nelms
 - “[Microelectronic Circuits](#)” by A. S. Sedra & K. C. Smith

Office hours are your friends, we get to know you better when you come to see us

Calendar

Week 1	2/27 L1: Introduction & Basic Concept	3/1 L2: Resistive Circuits
Week 2	3/5 L3: Circuit Analysis Techniques	3/8 L4: Capacitors & Inductors
Week 3	3/12 TA office hr	3/15 L5: Transient circuit analysis
Week 4	3/19 L6: AC steady-state analysis	3/22 L7: Frequency domain analysis
Week 5	3/26 L8: Two-port network	3/29 TA office hr
Week 6	4/2 Midterm Exam 1	4/7 L9: pn Junction & Diodes (1)
Week 7	4/9 L10: pn Junction & Diodes (2)	4/12 L11: Bipolar Junction Transistors (1)
Week 8	4/16 L12: Bipolar Junction Transistors (2)	4/19 L13: MOS Field-Effect Transistors (1)
Week 9	4/23 TA office hr	4/26 L14: MOS Field-Effect Transistors (2)
Week 11	5/7 TA office hr	5/10 Midterm Exam 2
Week 12	5/14 L15: Digital Integrated Circuits	5/17 L16: Operational amplifier & Feedback (1)
Week 13	5/21 TA office hr	
Week 14	5/28 L17: Operational amplifier & Feedback (2)	5/31 L18: Oscillators & Signal Generators (1)
Week 15	6/4 L19: Oscillators & Signal Generators (2) & Wrap up	
Week 16	6/11 TA office hr	

3 classes per lecture session

Attendance & Quiz

- Attendance

- Optional
- Slides posted online

- Quiz

- 5-10min Quiz by the end of **every** lecture session
- Lowest 3 quizzes dropped in final grading

Review the slides RIGHT AFTER every lecture
DO NOT wait until one week before the exams

Homework

- Assigned every week after lecture
- HW submission after due can get 50% score
- Discussion between classmates are encouraged
- Zero tolerance to cheating/plagiarism. Penalty for cheating/plagiarism is failure of the course, and will be turned over to the department

Grading

- Homework & Quiz – 20%
- Middle exam – 20% each
 - No makeup
 - Absences: 0
- Final exam – 40%

General philosophy

- Practice makes master

- Circuit is difficult
- The only way to get good is by practicing

- Self learning is important

- Not everything you need for homework is covered by my lecture
- Your teacher is a guide, not a spoon feeder

- You get what you put into this course

- I have no desire to force you to study

Who study circuit and systems?

Fundamentals of Electronic Circuits and Systems

CAS



**IEEE Circuit and
System Society**



**IEEE Solid-State
Circuit Society**

International societies are where you find your peers



The Prologue of CAS History

Fundamentals of **Electronic** Circuits and Systems



A. Volta

1800

Invention of
Voltaic Pile

CAS

The Prologue of CAS History

Fundamentals of **Electronic** Circuits and Systems



1800

Invention of Voltaic Pile



A.-M. Ampere



J. P. Joule



C. A. Coulomb



M. Faraday

CAS

The Golden Epoch

Late 1800s to 1900s

The Prologue of CAS History

Fundamentals of **Electronic** Circuits and Systems



1800

Invention of Voltaic Pile

CAS

The Golden Epoch

Late 1800s to 1900s



1870s

2nd industrial
revolution



A.-M. Ampere



J. P. Joule



C. A. Coulomb



M. Faraday

Course Objectives

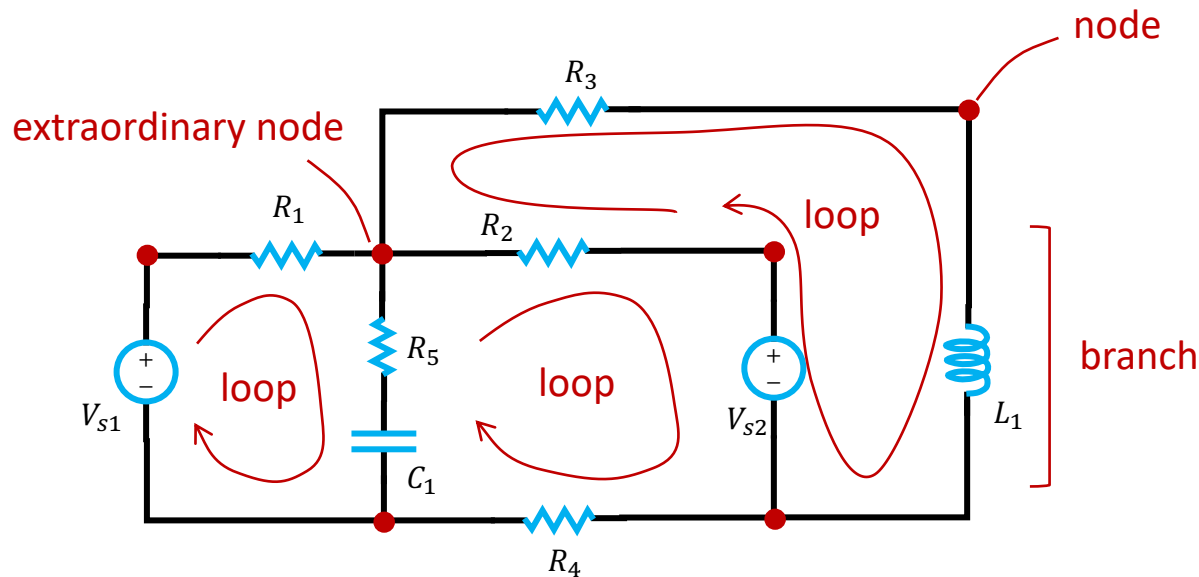
Fundamentals of Electronic Circuits and Systems

- **Devices:** Individual circuit elements
 - R/L/C/Diode/Transistors
 - Well understand and use the **I-V relationship**
- **Circuits:** can be considered as subsystems
 - Know the **function** of some classical circuits
 - Well understand the **analysis skill**
- **Systems**
 - Know some important systems

**To learn how to analyze the I-V relationship of
a device / a circuit / a system**

Circuit Topology

- **Branch**: signal element, such as a resistor or source
- **Node**: connection point between two or more branches
- **Extraordinary node**: connection point between at least 3 branches
- **Loop**: closed path in a circuit

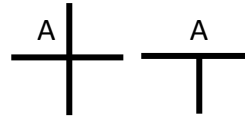


Common circuit elements

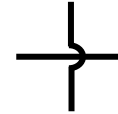
Connections



Conductor



Two conductors
electrically
joined at node A



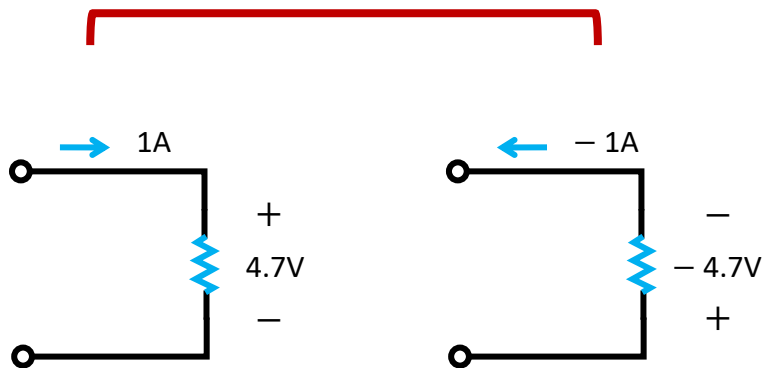
Two conductors
not joined
electrically



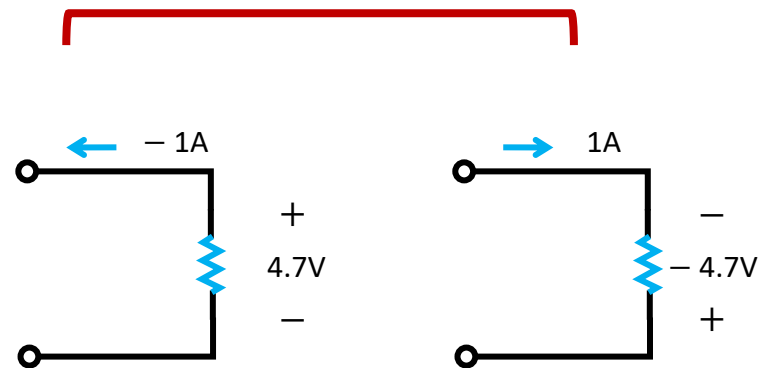
Switch

Associated variables convention

Positive current flow



Negative current flow



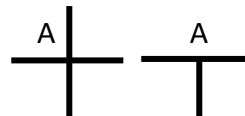
Associated Variables Convention define current to flow in at the device terminal assigned to be positive in voltage

Common circuit elements

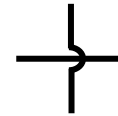
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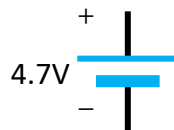


Two conductors
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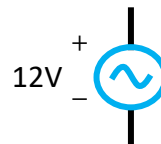


Switch

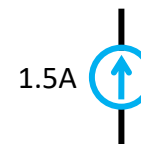
Supplies



4.7V DC



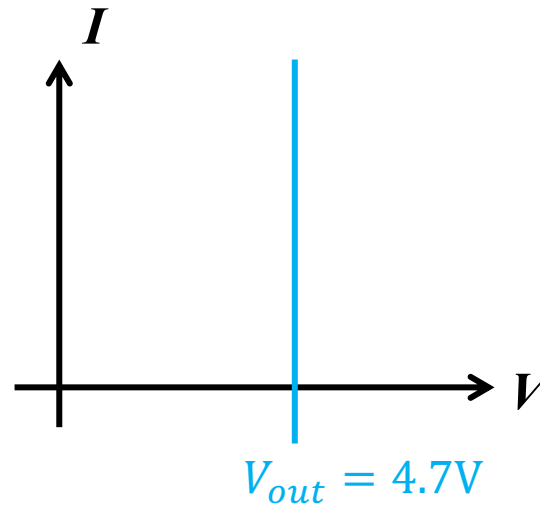
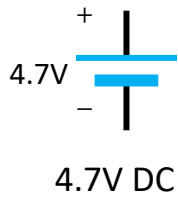
12V AC



1.5A current

I-V relationships of a voltage source

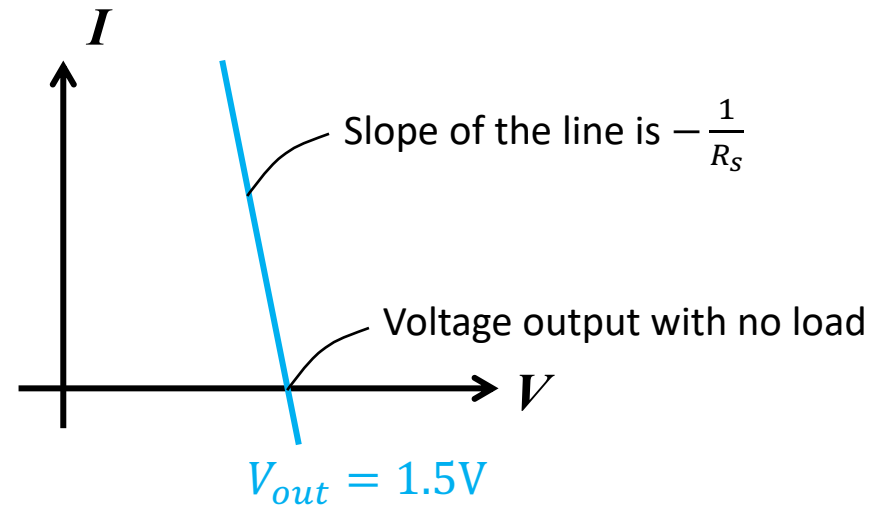
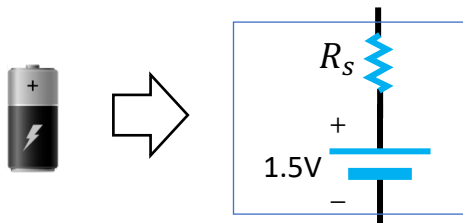
Ideal linear independent voltage source



- Provide a **FIXED** voltage output
- Provide **UNLIMITED** current output

I-V relationships of a voltage source

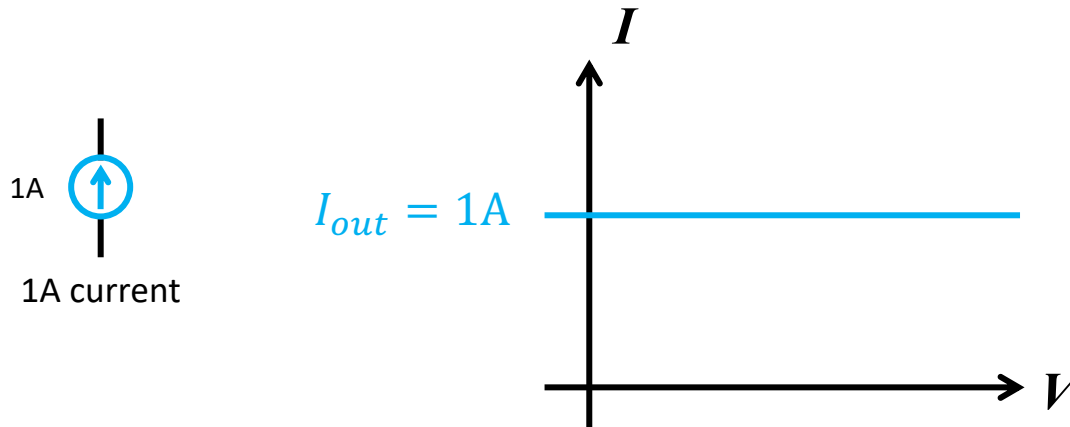
Practical voltage source



- Can **NOT** Provide a fixed voltage output
- There is an **internal resistance** in the model

I-V relationships of a current source

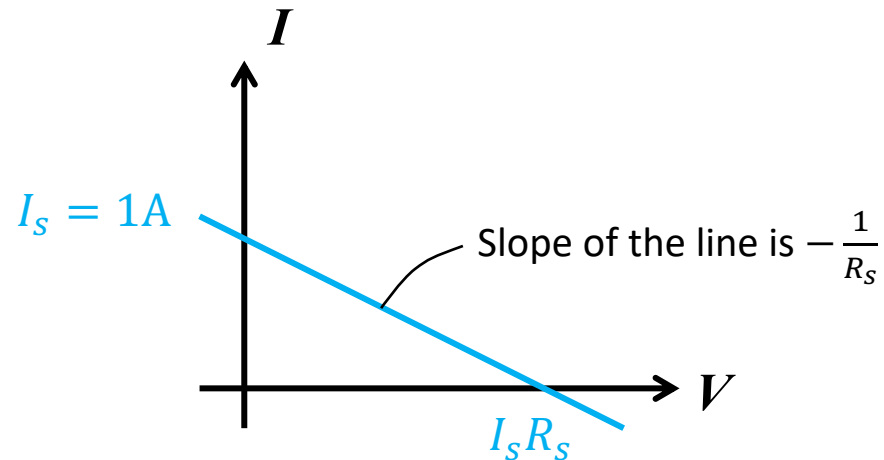
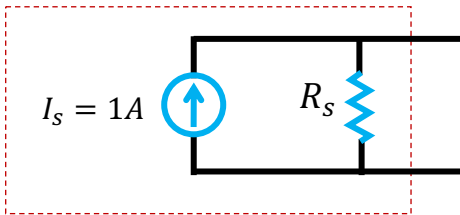
Ideal linear independent current source



- Provide a **FIXED** current output
- Provide **UNLIMITED** voltage output

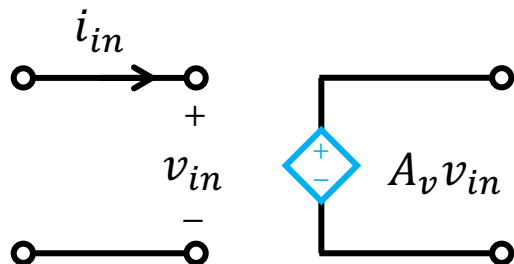
I-V relationships of a current source

Practical current source

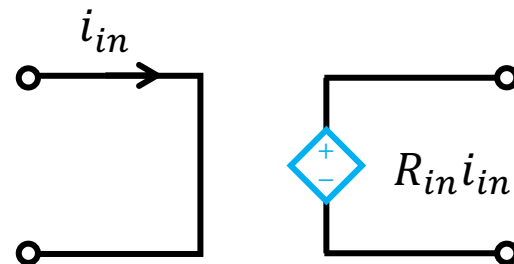


- Can **NOT** Provide a fixed current output
- There is an **internal resistance** in the model

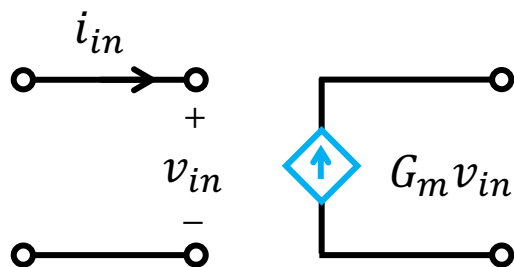
Linear **dependent** sources



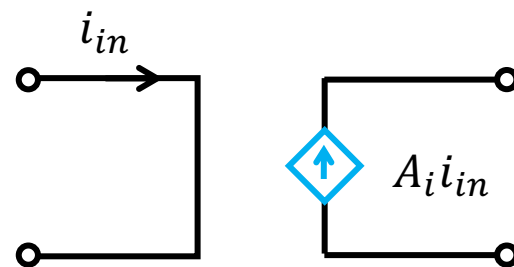
Voltage-controlled voltage source (VCVS)



Current-controlled voltage source (CCVS)



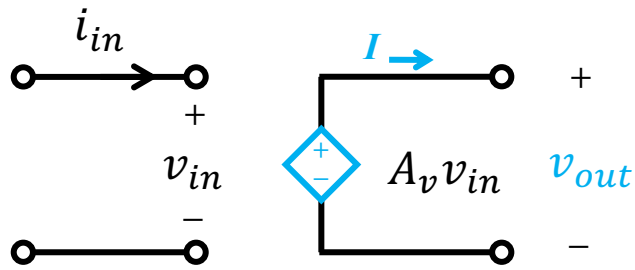
Voltage-controlled current source (VCCS)



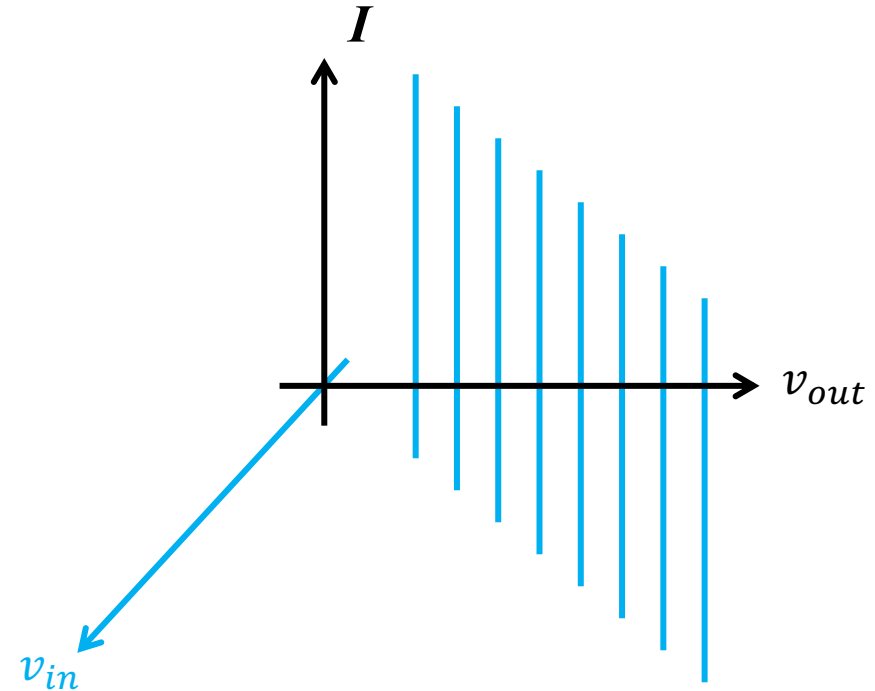
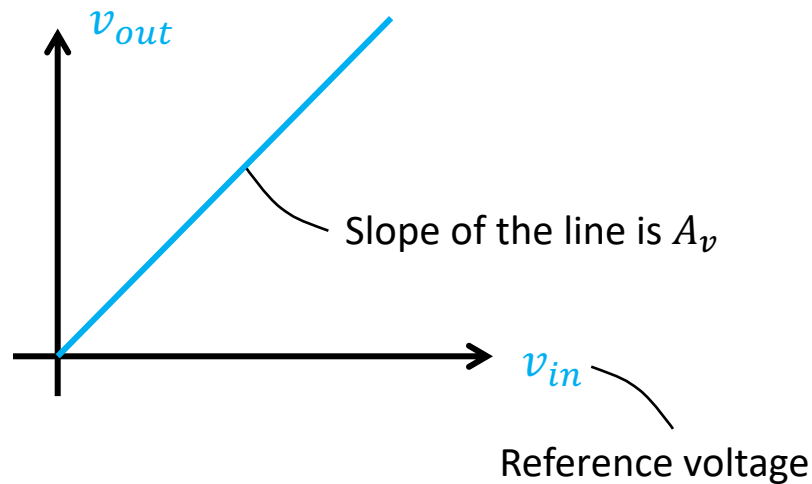
Current-controlled current source (CCCS)

The source voltage/current **DEPENDS** on a voltage/current elsewhere in the circuit

I-V relationship of dependent source



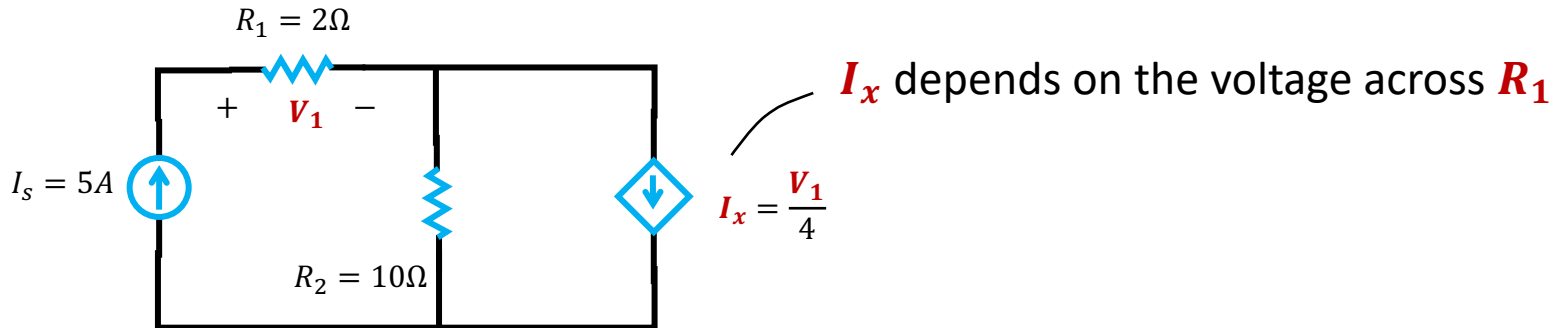
Voltage-controlled voltage source (VCVS)



**Dependent source output
varies with the reference
voltage/current**

Example 1

QUESTION: Find I_x in the circuit below:



SOLUTION:

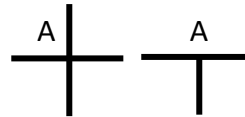
- According to **Ohm's Law** $V_1 = I_s R_1 = 10V$
- $I_x = \frac{V_1}{4} = \mathbf{2.5A}$

Common circuit elements

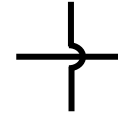
Connections



Conductor



Two conductors
electrically
joined at node A

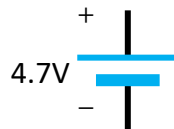


Two conductors
not joined
electrically

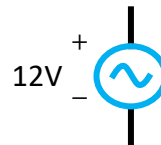


Switch

Supplies



4.7V DC



12V AC

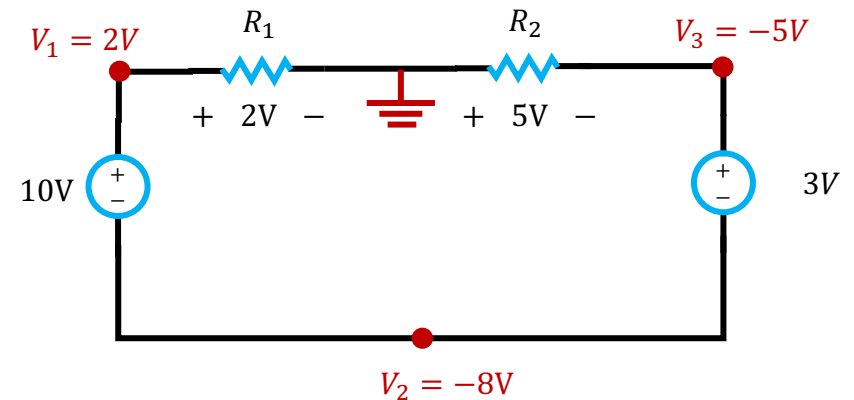
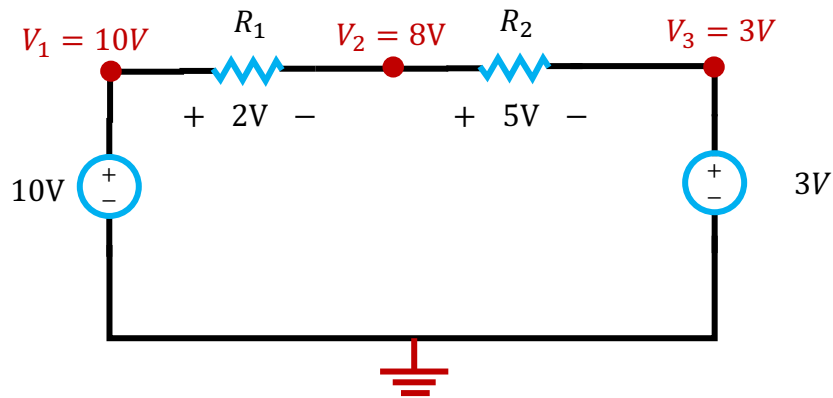


1.5A current



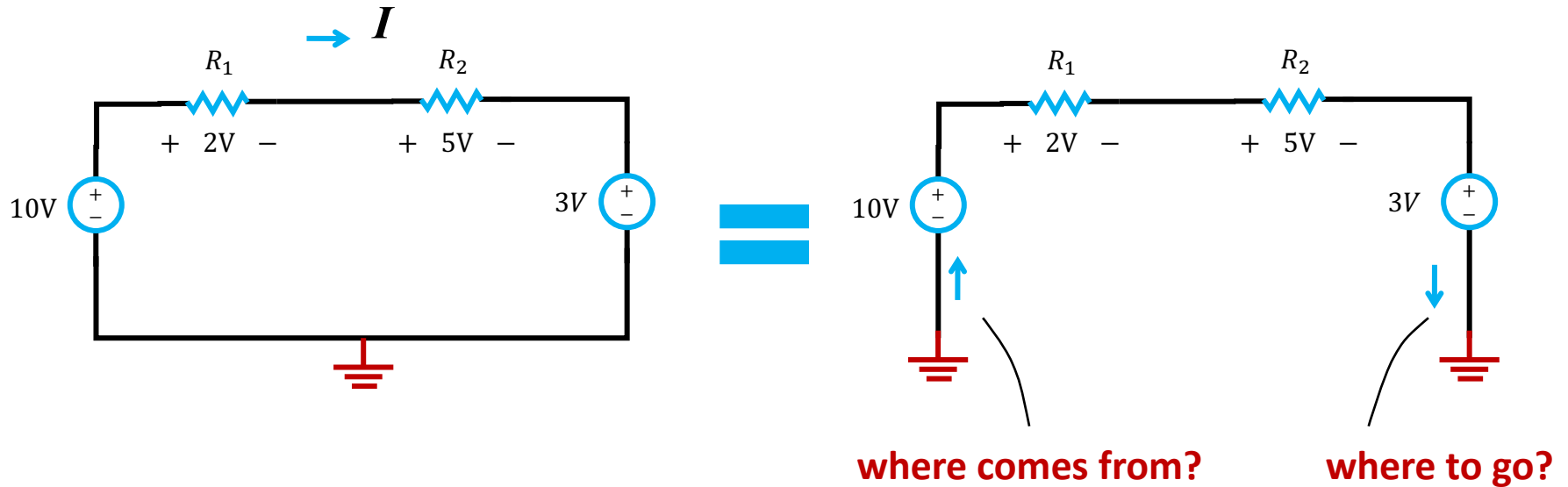
Voltage reference
(ground)

Reference / Ground



- Choose **REFERENCE** point for all the potentials
- Assign the potential at reference equals **0**, called **GROUND**
- All potentials are relative to ground terminal

Reference / Ground



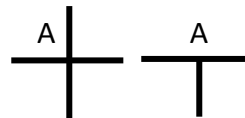
- Are these two circuits equal?
-  is only a symbol, **NOT** a branch

Common circuit elements

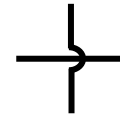
Connections



Conductor



Two conductors
electrically
joined at node A

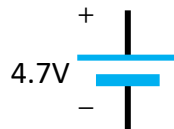


Two conductors
not joined
electrically

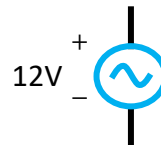


Switch

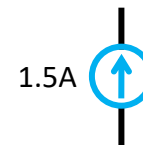
Supplies



4.7V DC



12V AC



1.5A current



Voltage reference
(ground)

Passive devices



Fixed value
resistor



Variable
resistor



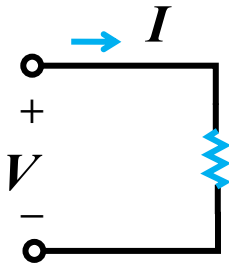
Capacitor



Inductor

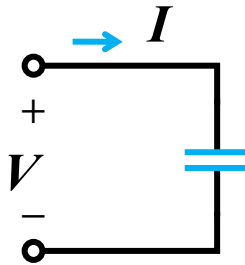
I-V relationship of passive devices

Resistor



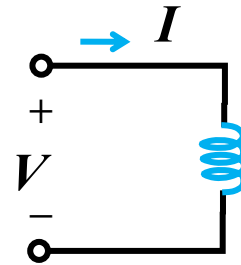
$$v(t) = R \cdot i(t)$$

Capacitor



$$i(t) = C \cdot \frac{dv(t)}{dt}$$

Inductor



$$v(t) = L \cdot \frac{di(t)}{dt}$$

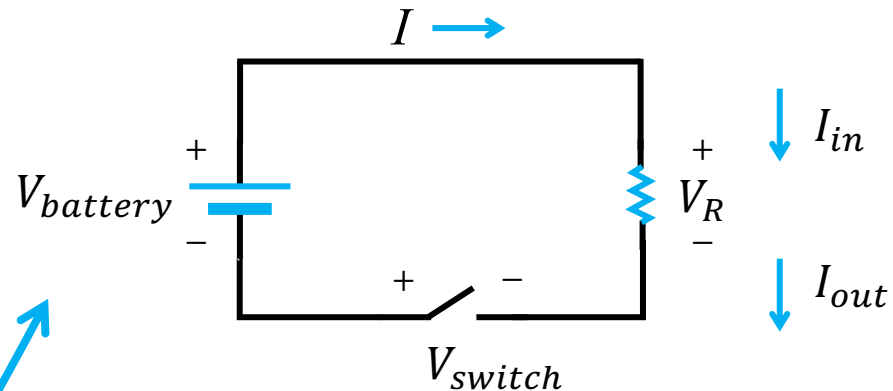
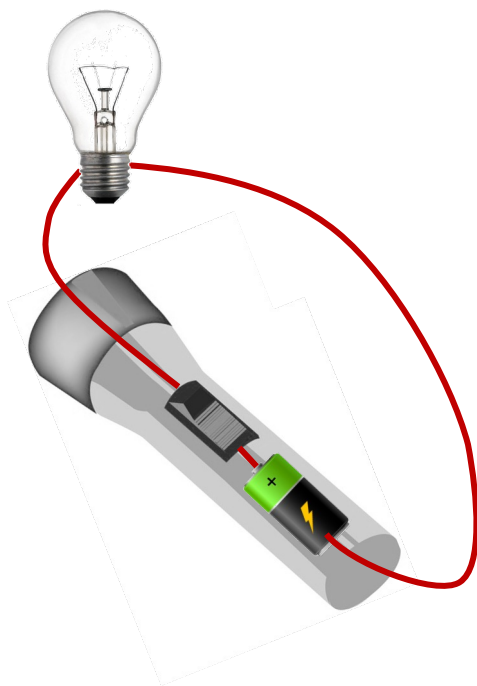
More details will be introduced in this semester

Outlines

- Welcome & course logistics/policies
- Course objectives
 - To learn how to analyze the **I-V relationship**
- Common **circuit elements**
 - A quick review
 - Dependent voltage/current sources
 - I-V relationship of voltage/current sources
 - I-V relationship of passive devices
- **An example of circuit**



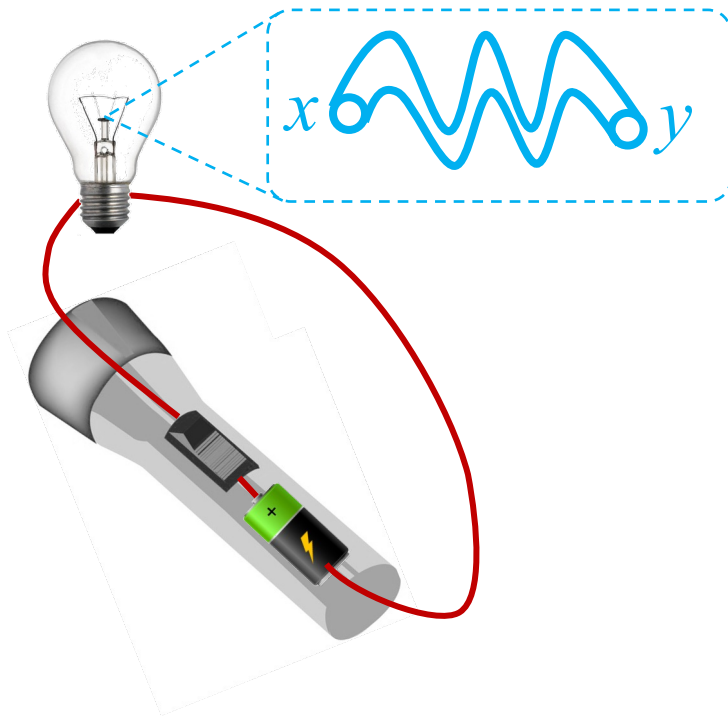
An example of circuit



What is the relationship between I and V_R ?

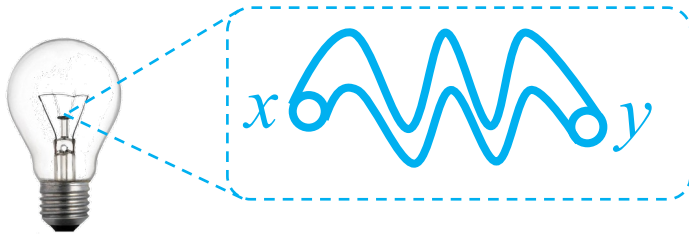
$$\left\{ \begin{array}{l} V_R = IR \\ V_R + V_{switch} = V_{battery} \\ I_{in} = I_{out} \end{array} \right.$$

According to Maxwell's equations



$$\left\{ \begin{array}{l} \int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} \\ \oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} \end{array} \right.$$

Lumped circuit elements



Real world scenario



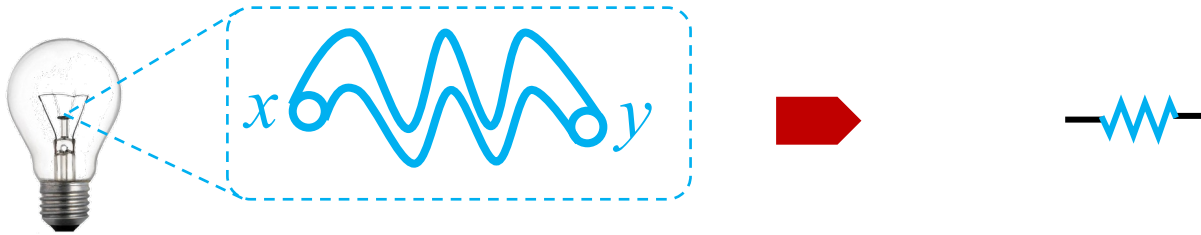
Lumped circuit elements

$$\left\{ \begin{array}{l} \int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} \\ \oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} \end{array} \right.$$

$$\left\{ \begin{array}{l} I_{in} = I_{out} \\ V_R + V_{switch} = V_{battery} \end{array} \right.$$

Are they the same?

Lumped Matter Discipline (LMD)



- The rate of change of magnetic flux linked with any portion of the circuit must be zero for all time.

$$\oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} = 0$$



$$V_R + V_{switch} = V_{battery}$$

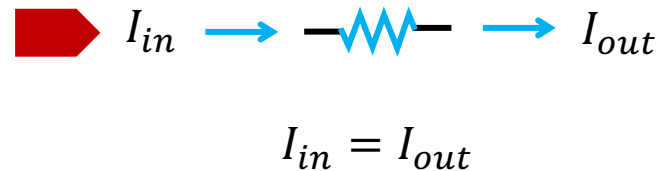
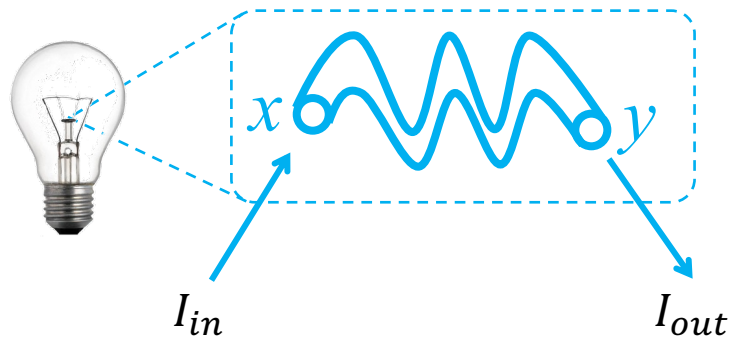
- The rate of change of the charge at any node in the circuit must be zero for all time. A node is any point in the circuit at which two or more element terminals are connected using wires.

$$\int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} = 0$$



$$I_{in} = I_{out}$$

Size issue



$$I_{in}(t) \quad \neq \quad I_{out}(t + \Delta t)$$

- What if the wire is very long?
- What if Δt is very large?

The signal timescales must be much larger than the propagation delay of electromagnetic waves through the circuit.

Examples of size issue

- @60Hz

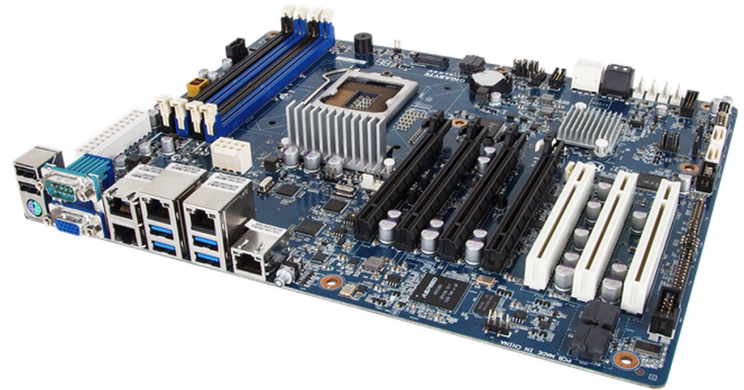
$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{60 \text{ Hz}} = 5000 \text{ km}$$



The resistance on the power line cannot be ignored

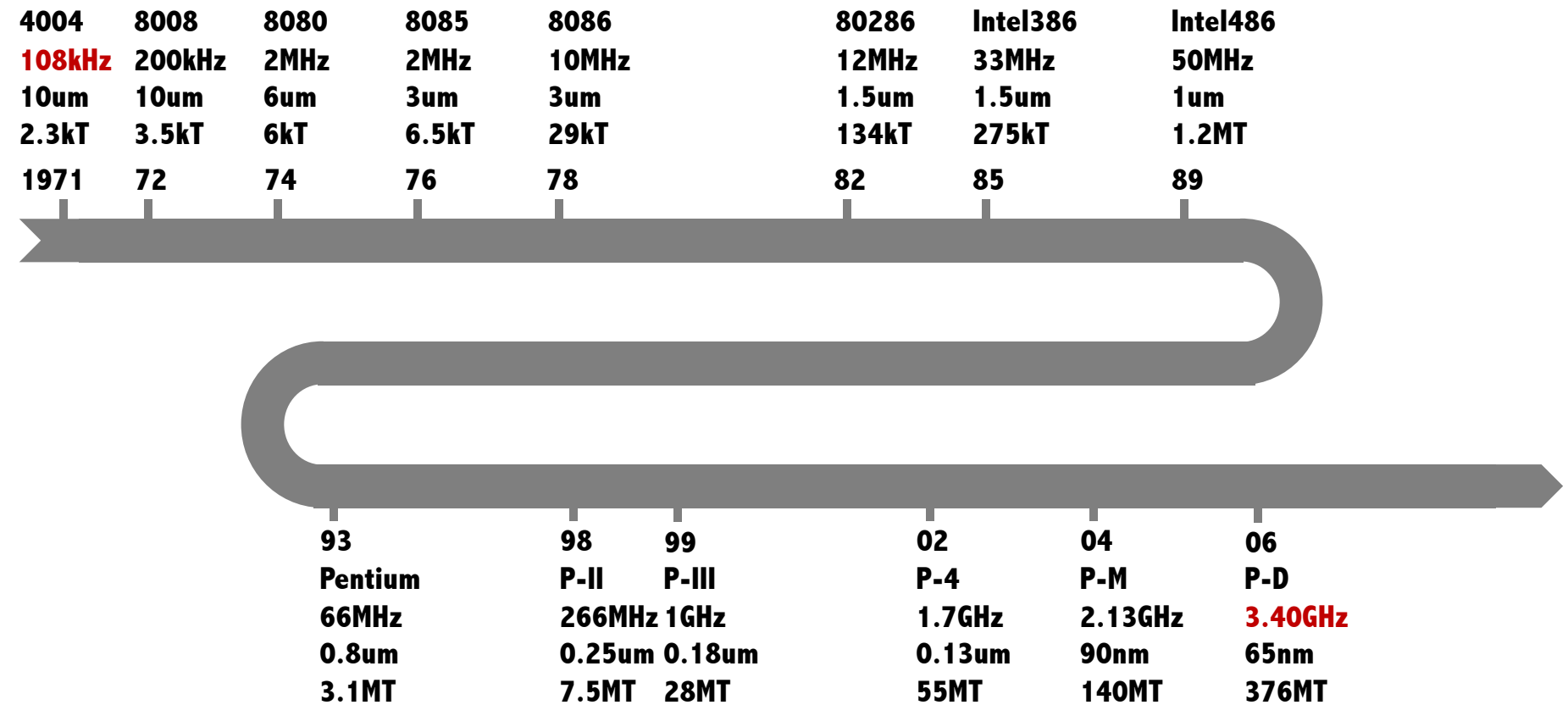
- @1GHz

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{1 \text{ GHz}} = 300 \text{ mm}$$



LMD cannot be simply applied for 30cm motherboard with 1GHz frequency

Examples of size issue



$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{108 \text{ kHz}} = 2.78 \text{ km}$$



$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{3.4 \text{ GHz}} = 88.2 \text{ mm}$$

Lumped Matter Discipline (LMD)

- The rate of change of magnetic flux linked with any portion of the circuit must be zero for all time.

$$\oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} = 0$$



$$V_R + V_{switch} = V_{battery}$$

- The rate of change of the charge at any node in the circuit must be zero for all time. A node is any point in the circuit at which two or more element terminals are connected using wires.

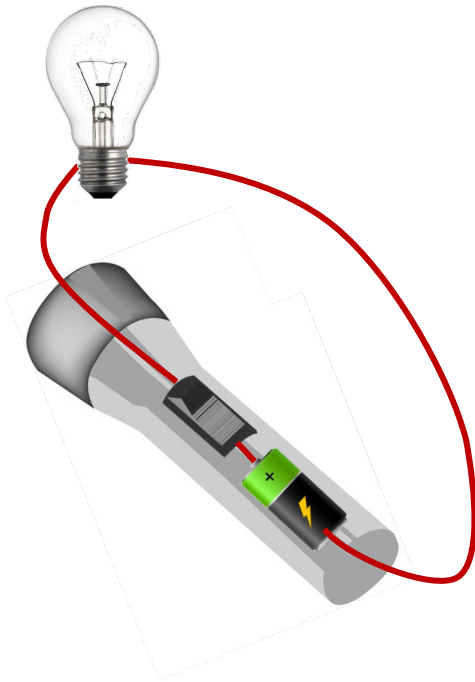
$$\int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} = 0$$



$$I_{in} = I_{out}$$

- The signal timescales must be much larger than the propagation delay of electromagnetic waves through the circuit.

WHY LMD?



- According to Maxwell's equations

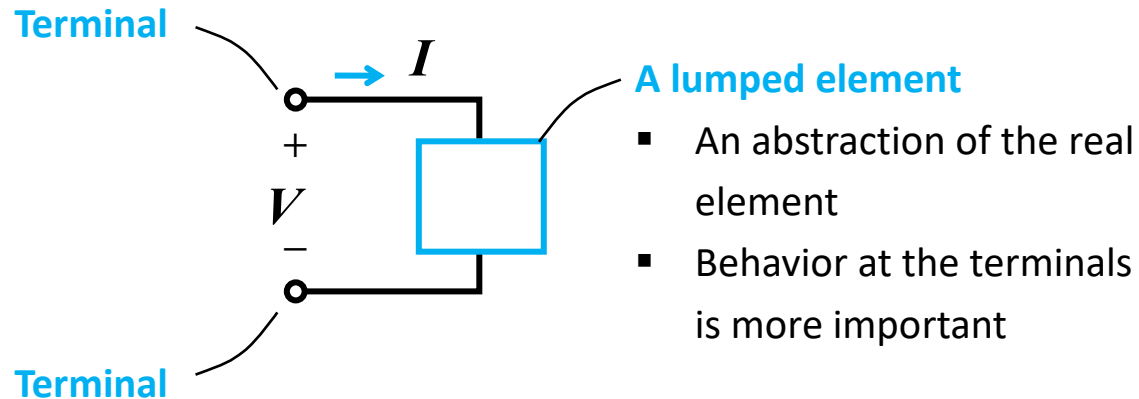
$$\begin{cases} \int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} \\ \oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} \end{cases}$$

- After applying LMD

$$\begin{cases} I_{in} = I_{out} \\ V_R + V_{switch} = V_{battery} \end{cases}$$

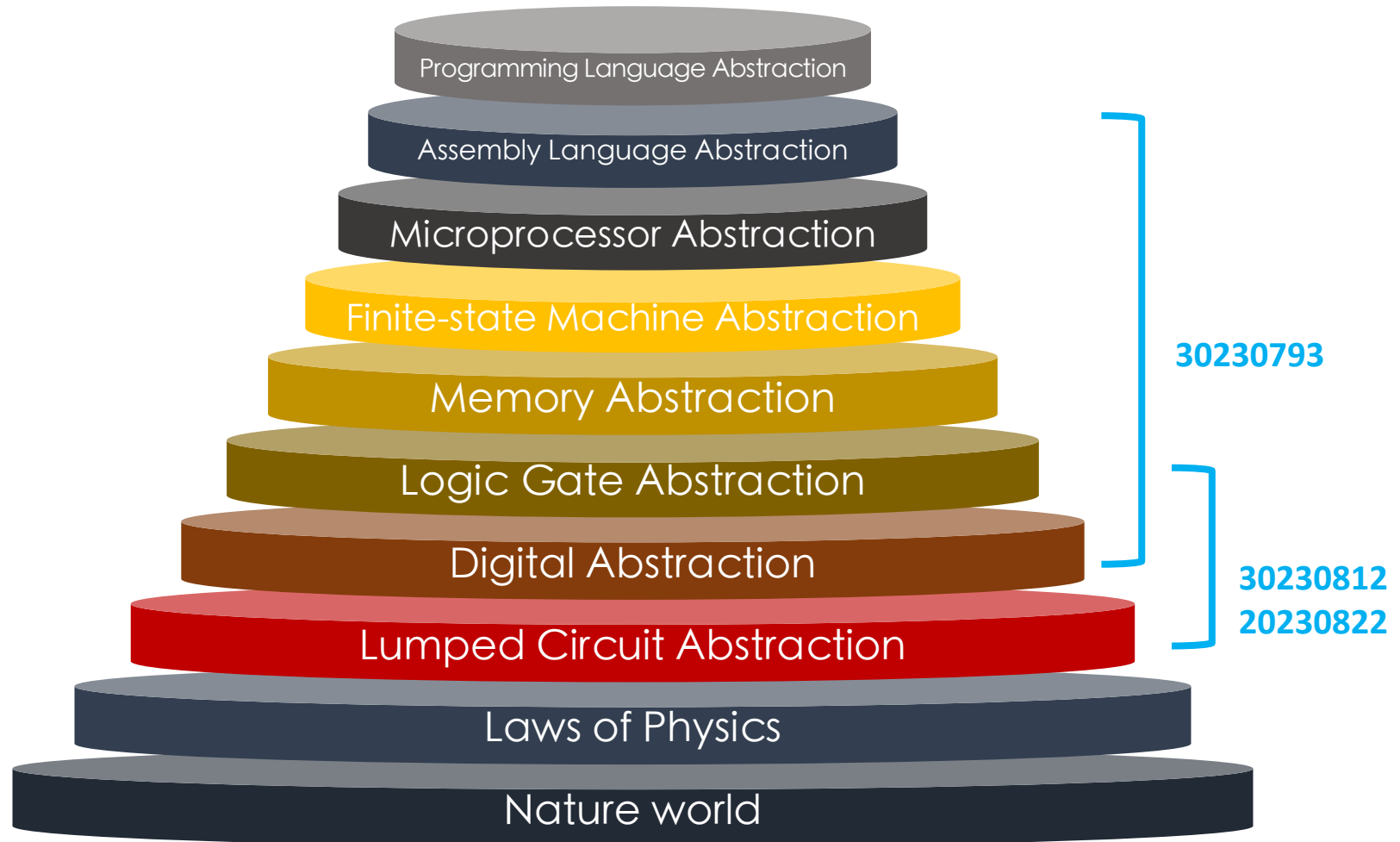
A dramatic reduction on computational load

Lumped circuit abstraction



The lumped circuit abstraction capped a set of lumped elements that obey the lumped matter discipline (LMD) using **ideal wires** to form an assembly that performs a specific function results in the lumped circuit abstraction.

Why Abstraction?

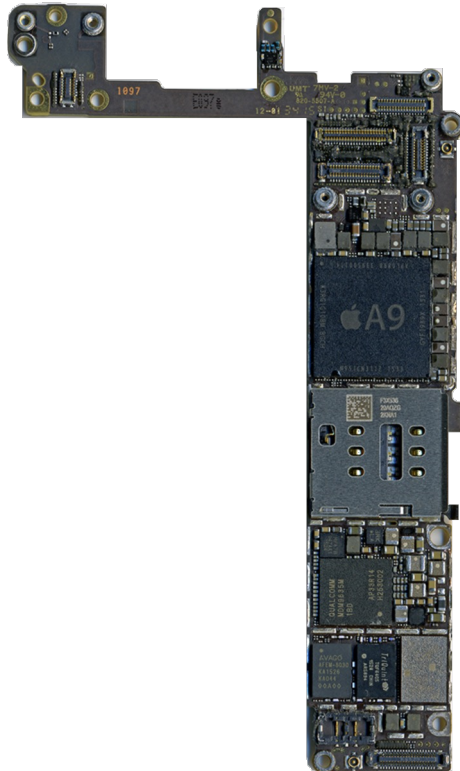


Abstraction is a pathway from the real world to engineering world

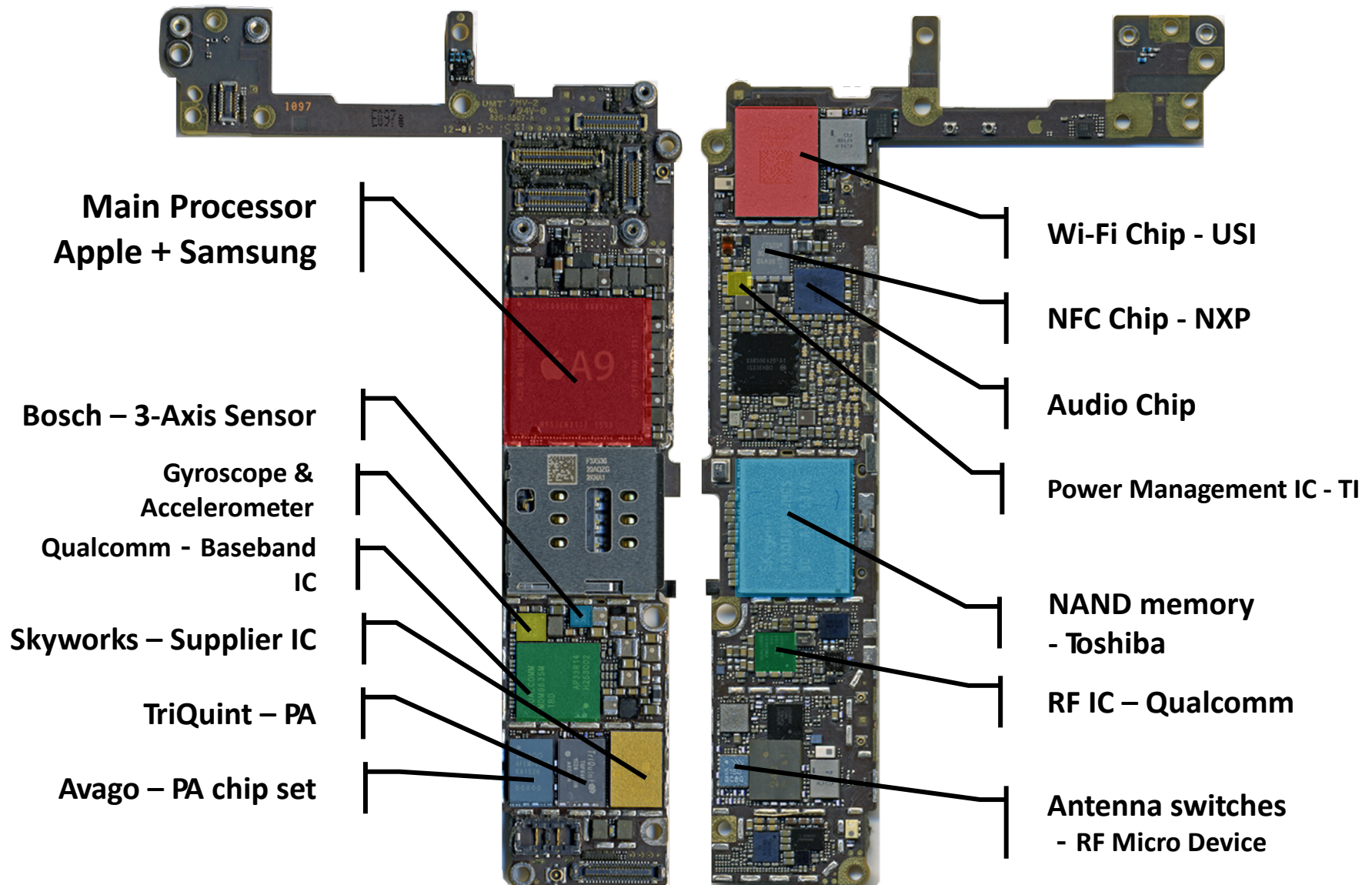
Outlines

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- Course objectives
 - To learn how to analyze the **I-V relationship**
- Common **circuit elements**
 - A quick review
 - Dependent voltage/current sources
 - I-V relationship of voltage/current sources
 - I-V relationship of passive devices
- An example of **circuit**
 - Lumped Matter Discipline (LMD)
- **An example of system**

An example of system



iPhone system



Architecture of an iPhone system



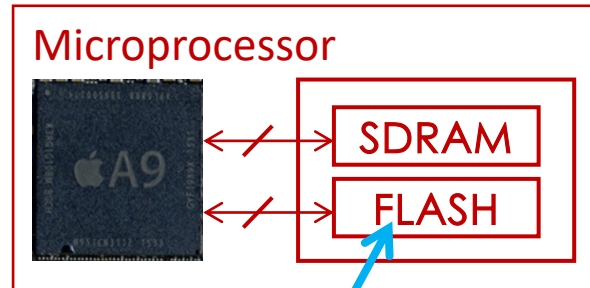
Microprocessor

Power module

Radio Frequency module

Human Interface

Architecture of an iPhone system



iPhone 6s Plus

SIM-free

Silver

Capacity

Now choose your capacity.

32GB¹

\$649.00

Delivery: In Stock

Pickup: [Check availability](#)

128GB¹

\$749.00

Delivery: In Stock

Pickup: [Check availability](#)

- SDRAM

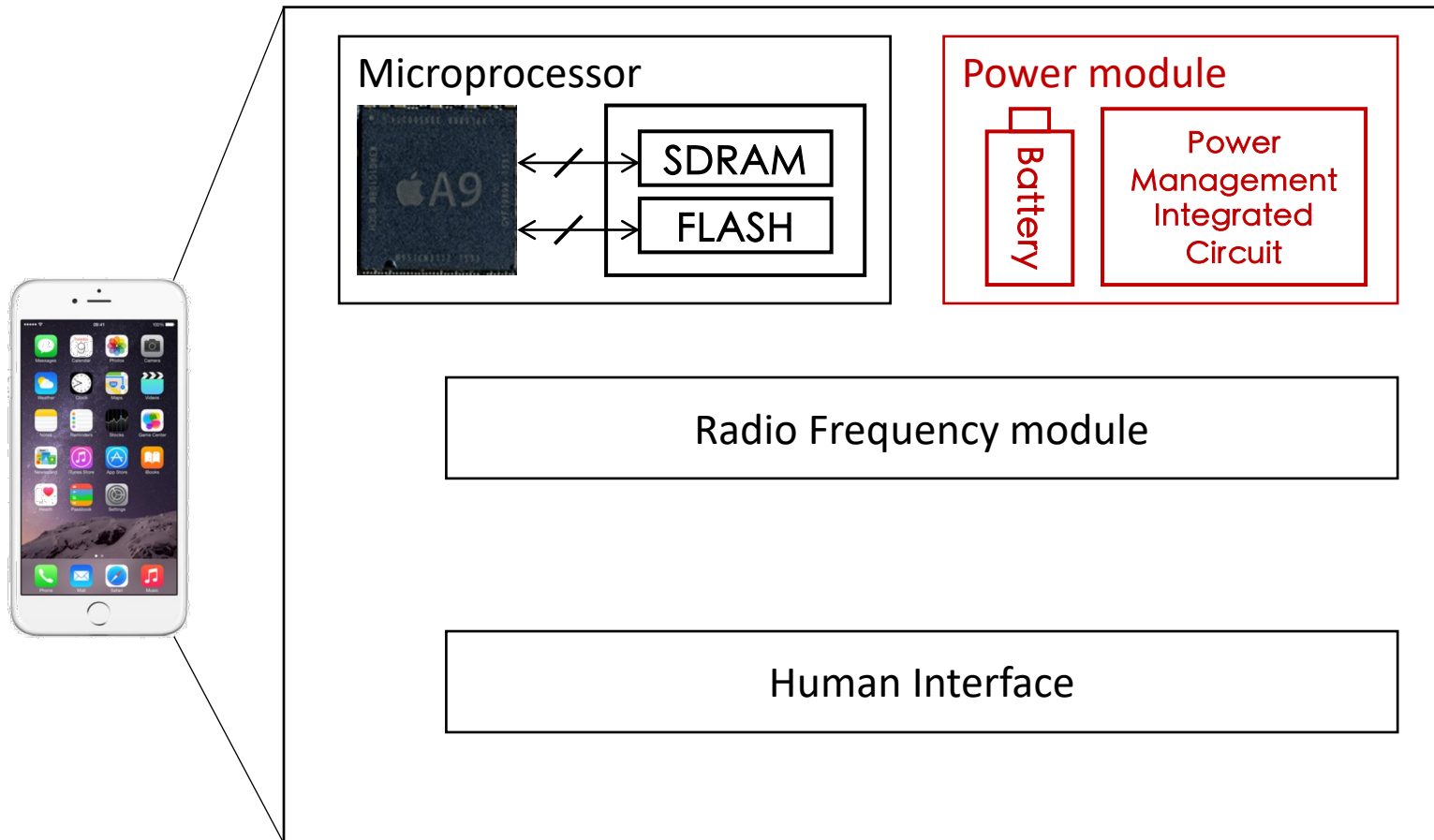
- i.e. DDR

- Flash

- EPROM, EEPROM

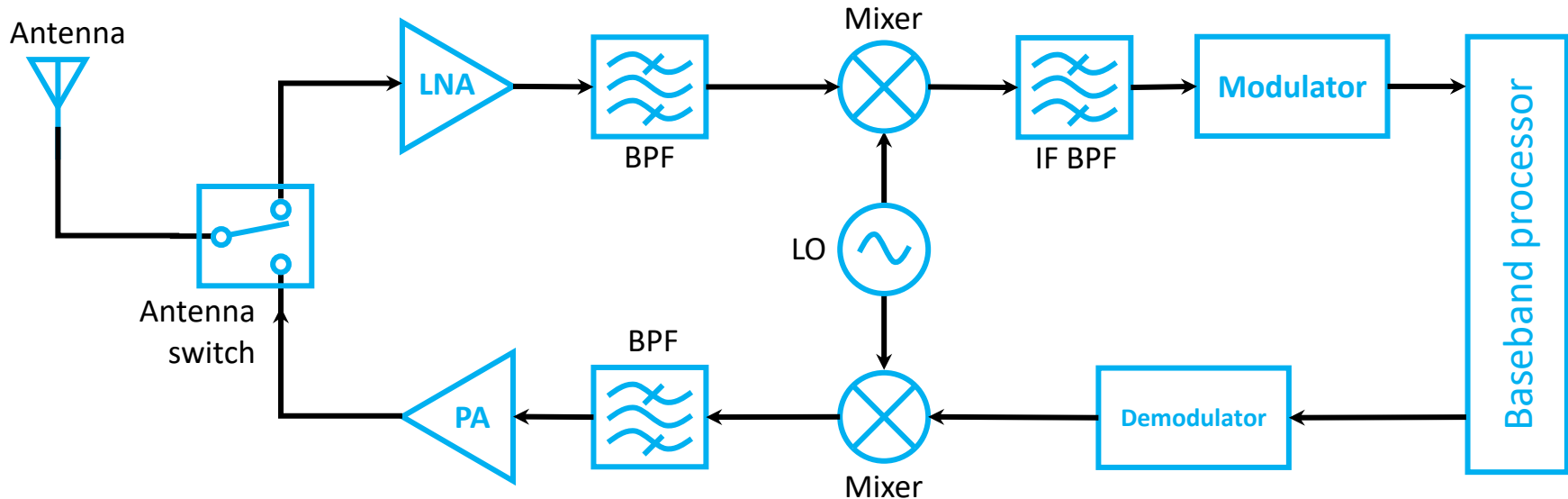
- NOR Flash, NAND Flash

Architecture of an iPhone system



Radio Frequency Module

Why we transfer signal at such a high frequency?



Human vocal speech frequency: 300Hz – 3.4kHz

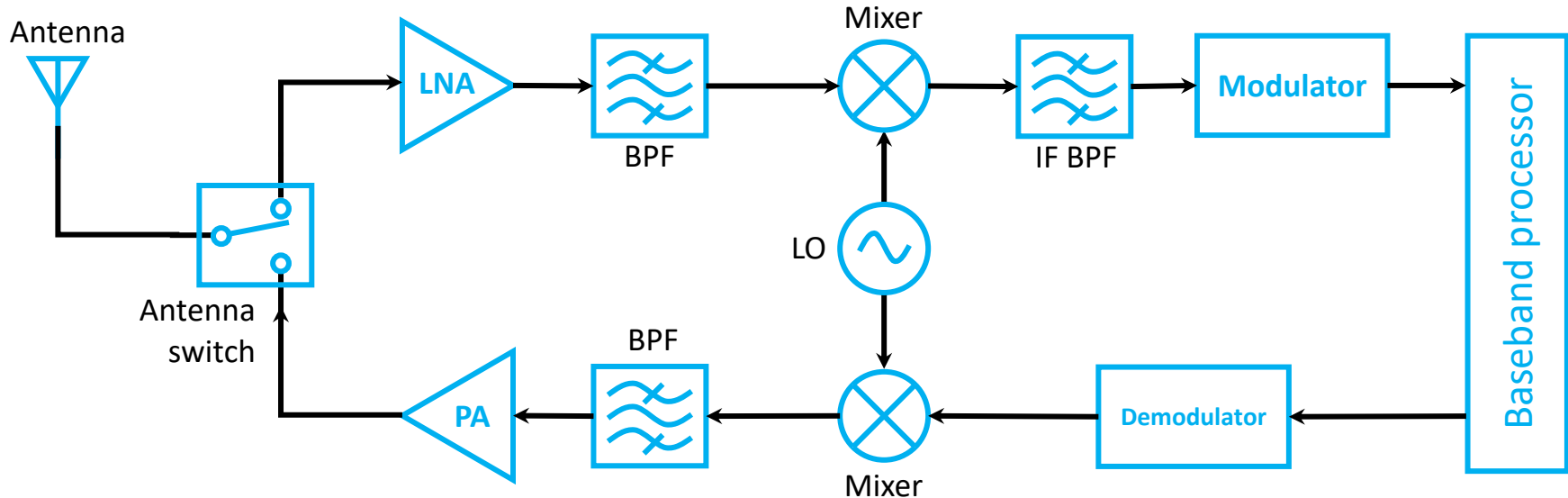
Frequency band for 4G

- China Mobile TDD-LTE: 2575MHz – 2635MHz
- China Unicom TDD-LTE: 2300MHz – 2320MHz
- China Telecom TDD-LTE: 2635MHz – 2655MHz
- China Unicom FDD-LTE: 1755 – 1765 / 1850 – 1860 MHz
- China Telecom FDD-LTE: 1765 – 1780 / 1860 – 1870 MHz

RF	Radio Frequency
BPF	Bandpass Filter
LNA	Low Noise Amplifier
PA	Power Amplifier
LO	Local Oscillator
Mixer	Frequency Up/Down Converter
IF	Intermediate Frequency
LTE	Telecommunication
TDD	Time Division Duplex
FDD	Frequency Division Duplex

Radio Frequency Module

Antenna size: $\frac{\lambda}{4}$

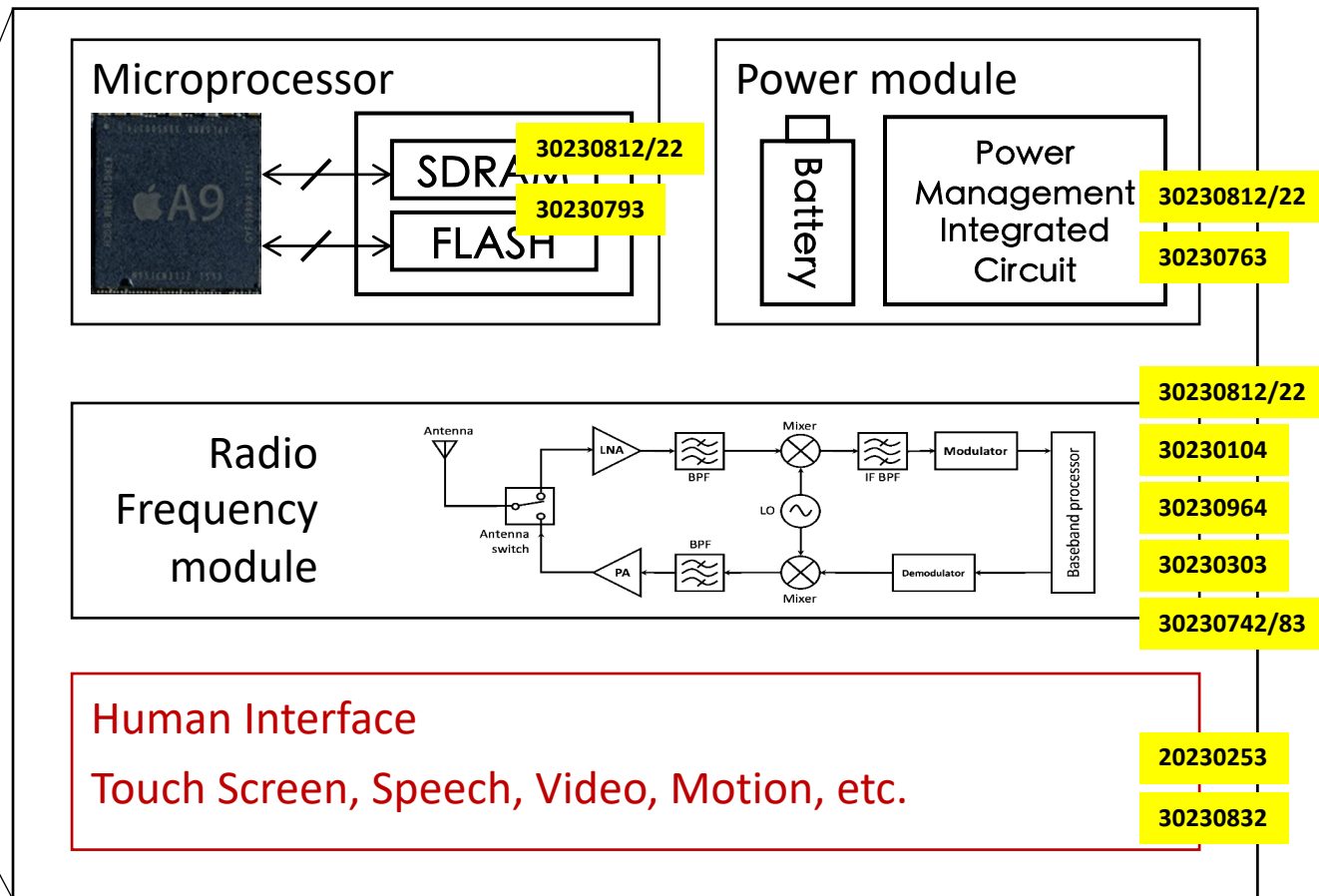


Wavelength of 1kHz Human vocal speech: $\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{1 \text{ kHz}} = 300 \text{ km}$

Wavelength of 2600MHz RF: $\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{2600 \text{ MHz}} = 11 \text{ cm}$

The RF module “moves” baseband signal to high frequency

Architecture of an iPhone system



Outlines

- Welcome & course logistics/policies
- Course objectives
 - To learn how to analyze the **I-V relationship**
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Reading tasks & learning goals

■ Reading tasks

- Basic Engineering Circuit Analysis
 - Chapter 1
- Foundations of Analog and Digital Electronic Circuits
 - Chapter 1 (optional)

■ Learning goals

- Understand the key of this course: **I/V relationship**
- Known the I/V relationship of the **power sources**
 - I/V relationship of the ideal linear independent voltage/current source
 - I/V relationship of the Practical voltage/current source
- Understand the Lumped Matter Discipline (LMD)
- Understand the lumped circuit abstraction